

We are leaving architecture mode.

The computational engine, editorial ladder, Claim Firewall, and Gate Firewall are frozen V1. Do not redesign infrastructure unless you hit a genuine implementation blocker. The product now evolves by adding features and polish, not by redesigning architecture.

STEP 0 — SMOKE TEST

Compute a real fingerprint for:

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Castellón, Spain
12 August 2026
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Print the complete facts payload.

IF THE SMOKE TEST FAILS, classify the failure before taking any action:

(A) INTEGRATION failure — the engine could not run to completion: import errors, missing dependencies, file-path or I/O errors, serialization/formatting bugs, a crashed service or endpoint. The astronomy was never in question; the plumbing broke. → Diagnose, repair, rerun the smoke test, and continue. This is normal build work.

(B) SCIENTIFIC failure — the engine ran to completion but a computed value is wrong or suspect: a date, coordinate, distance, duration, Sun altitude, in-umbra result, Saros number, or any figure a user could read as fact. → STOP. Do not modify the engine, its constants, its ΔT model, its data, or its outputs. Print the payload, state which value is suspect and why, and report for human review. The engine is frozen V1; changing a number is a human-gated event, never an autonomous repair.

IF YOU CANNOT CONFIDENTLY CLASSIFY THE FAILURE AS (A), TREAT IT AS (B) AND STOP. When in doubt, the engine is sacred.

THE SMOKE TEST SUCCEEDS — BUILD

Implement the next production increment of the complete end-to-end experience, using only real engine output:

1. birthplace
2. birthdate
3. compute

4. reveal (the animated moment – fast, punchy, ~2-3s, replayable)
5. scientific verdict
6. shareable Eclipse Fingerprint (the verdict card – the growth engine; build it first-class, server-rendered from the deterministic facts so the shared image pixel-identical to what the user saw)
7. future-eclipse invitation – ONLY when next_totality_home is non-null. If it is this screen does not exist; end the journey beautifully on the card. Never show apologetic empty state.

RULES • No mock data. No placeholder copy. If a fact is null, omit it — never fabricate a fallback. • Deterministic engine only. Same input → same output → same choreography, always. • All narration passes through the Claim Firewall. If a sentence fails, drop to the deterministic headline, and make that headline-only state look intentional — for some users it is what ships. • Every screen production-ready.

PRIORITIES (break ties with the first) 1. emotional impact 2. clarity 3. speed 4. beauty 5. scientific credibility

Whenever multiple UX solutions exist, choose the version most emotionally memorable and most likely to be shared. The emotional payload is TIME and RARITY, not proximity — dramatize whatever the frozen headline ladder already chose as the winner; never invent a fact, only pace the reveal of one.

Think like a Lead Product Designer, Creative Director, and Senior Front-End Engineer working together. Do not stop after planning. Implement the next production increment.